Data sources

Above: ARCHITECTURE.md §4.7 (outside the box).

Talks to: every block that needs the outside world. Each one reaches it through adapters:

Block	Sources it uses
the desk	the scanner, quotes, short interest, borrow, news, index and sector quotes, outcome bars
the research worker	EDGAR, 13F filings, Polygon, index bars
qp	chart bars
the journal	account fills, bars

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1. What it is

Every outside source the desk uses, and the **adapter** in front of each one.

An adapter is the only code that talks to one source. A provider that offers several kinds of data, such as Yahoo (quotes, charts) or Alpaca (bars, borrow), has one adapter for each kind. Each adapter keeps in one place everything about its source:

- how to ask;
- how fast it may be asked;
- how fresh its answers stay;
- how to read them;
- what "nothing" and "could not ask" mean;
- how its health is judged.

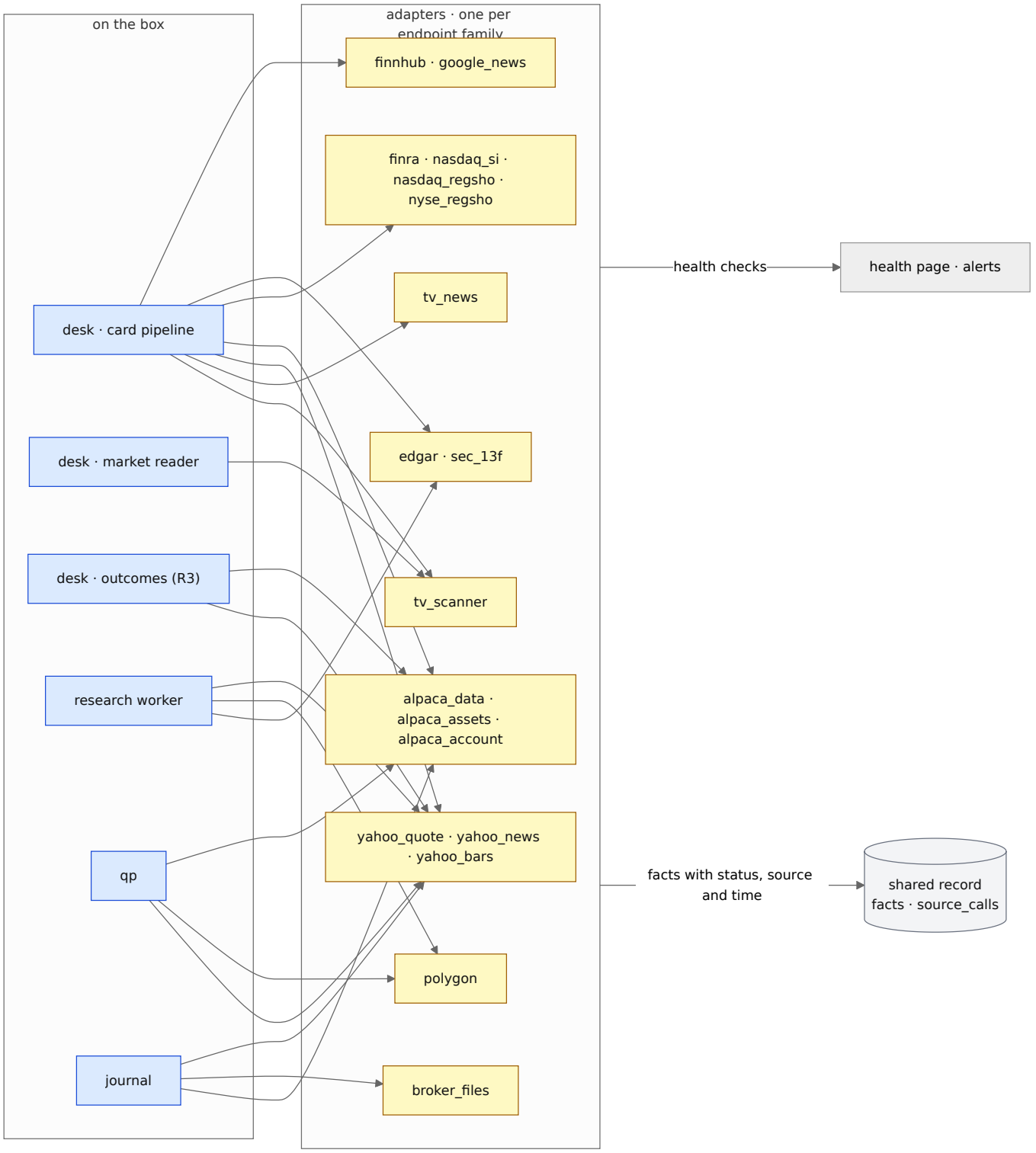
Every other block asks adapters for **facts**, and never calls a source directly.

What the adapters never do:

- guess a missing value (a missing value is `unavailable`, with the reason);
- read a missing value as 0, as "no" or as bearish;
- let a second source overwrite the first;
- let a source's failure fail a card.

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2. The diagram



3. Its parts

3.1 The adapter contract

Each adapter states	Meaning
families	the fact families it can produce
request	endpoint, method, headers, timeout
auth	none, a key (named in the secret file), or a User-Agent
limits	requests per second or per day; batch size; time budget
freshness	how long a fact it produced stays fresh, per family
parse	turning the answer into fact values, with alignment and sanity checks
honest null	what `none` means (the source answered: nothing) and what `unavailable` means (could not ask), each with a reason
health	how its content is judged: ok, degraded, down, on a schedule
fixtures	recorded answers, including failures, for CI

Every call is counted. Each fact family keeps a count of the requests it received, the calls it actually made, its failures and its last error, per day (`source_calls`). These counts decide whether sharing a family across tools pays (SHARED-RECORD.md).

3.2 Every source

Adapter	Endpoint(s)	Auth	Families	Quirks
tv_scanner	`POST scanner.tradingview.com/america/scan?label-product=screener-stock`	none; a browser User-Agent, Origin and Referer of tradingview.com; 15 s	candidates per screener; `quote` (32 columns, batches of 200); index and sector-ETF quotes	**ignores unknown columns silently**, so the column list is verified. A batch whose rows differ in length from the columns asked is refused (D-S3). It warns when two range windows are identical on most rows. `short_percentage_of_float` is never filled. Longer-window highs are not split-adjusted
tv_news	`news-headlines.tradingview.com/v2/headlines`	none	`news`	—
yahoo_quote	`quoteSummary`	cookie and crumb from `fc.yahoo.com`	`short_interest` (fallback)	per symbol
yahoo_news	`query1.finance.yahoo.com/v1/finance/search`	none	`news`	—
yahoo_bars	chart API (1-minute, daily), with or without extended hours	none	bars	shallow 1-minute history (weeks); years of daily
alpaca_data	`data.alpaca.markets/v2/stocks/bars` , `/v1beta1/news`	key	bars (fallback), `news` (Benzinga)	the free plan's IEX feed: thin volume, about a week of intraday history, regular hours only
alpaca_assets	the assets API	key	`borrow` (shortable, easy to borrow)	**as of now only**, with no history: a backtest applies today's flag and says so
alpaca_account	`GET /v2/account` , `GET /v2/account/activities/FILL` (paper or live)	each journal account's own key, read-only	journal fills; key test	at most 50 pages × 100 fills per fetch
finra	`cdn.finra.org/equity/regsho/monthly/shrt{date}.txt` , then the OTC biweekly paths	none	`short_interest` (shares short; the card works out the % of float)	**one file for every symbol**, fetched once a day; settlement twice a month
nasdaq_si	`api.nasdaq.com/api/quote/{t}/short-interest`	none	`short_interest` (fallback, two weeks of history)	per symbol
nasdaq_regsho	`nasdaqth{yyyymmdd}.txt`	none	`reg_sho`	per settlement day
nyse_regsho	the NYSE threshold-securities download	none	`reg_sho`	per settlement day
edgar	`data.sec.gov/api/xbrl/companyfacts/CIK{…}.json` ; `www.sec.gov/files/company_tickers.json` ; submissions (SIC, 8-K items); full-text search	a descriptive User-Agent (403 without it)	`oneil_c` , `oneil_a` , shares for `oneil_s` , `sic` , 8-K events, `news`	ten requests a second; the five EDGAR traps (ONEIL.md); filings land weeks after the quarter's end
sec_13f	the SEC's quarterly Form 13F datasets	User-Agent	`oneil_i`	quarterly, 45 days after the quarter's end; managers over $100M only
polygon	grouped daily (the whole market); aggregates	key	`bars_universe` , bars	the free plan is end of day: no bars for today
finnhub	`finnhub.io/api/v1/company-news`	key	`news`	—
google_news	`news.google.com/rss/search`	none	`news`	asked only when the others return fewer than 2 items
openfda, **usaspending**	drug and device approvals; contract awards	none	N events (planned)	—
broker_files	uploaded TradingView, TTP and DXtrade files	—	journal fills	one parser per format, each with a sample file (IMPORT.md)

3.3 Fallback orders

Family	Order	Rule
`quote`	TradingView	a misaligned batch is refused; the last quote stays, with its age
`short_interest`	FINRA → Yahoo → Nasdaq	stop at the first plausible answer and record which source gave it. With no float, the % is of shares outstanding, and it says so
`borrow`	Alpaca	`unavailable` never means "not shortable"
`reg_sho`	Nasdaq + NYSE	both are asked; either listing counts
`news`	TradingView, Alpaca, Yahoo, EDGAR and Finnhub in parallel; Google when fewer than 2 items	the failing ones are named; the others still count
outcome bars	Yahoo → Alpaca	per ticker
chart bars	the chosen feed	never silently replaced (FEEDS.md)
index bars (M)	Yahoo	—
whole-market bars (RS)	Polygon	—

A second source for the same fact is stored beside the first, with the priority recorded. It never overwrites the first (the F-21 lesson).

3.4 Bars

Every consumer gets the same shape: the chart, strategies, backtests, alerts, outcomes and the journal all receive a UTC time-indexed frame of open, high, low, close and volume.

Feed	Source	Strength	Weakness
`yahoo`	Yahoo	free, current, years of daily	shallow 1-minute history
`yahoo_ext`	Yahoo with pre- and after-market	one source, more hours	—
`polygon`	Polygon	deep history, extended hours, consolidated tape	free plan: no bars for today
`alpaca`	Alpaca IEX	live	thin IEX volume, short intraday history, regular hours
`hybrid`	Polygon history + Alpaca today	deep and today	**a volume seam**: IEX volume is a fraction of Polygon's, which distorts every VWAP on the Alpaca part
`hybrid_yahoo`	Polygon history + Yahoo today	deep and today, both consolidated	—

Where the feed detail lives:

- choosing the feed, warm-up and the bar cache with its limits: FEEDS.md;
- outcome bars, stored once per ticker per day and shared by every tool: REGISTERS.md.

3.5 Health checks

The data check of the old desk asked each source for data that **cannot** legitimately be empty, and judged the content, but only when a button was pressed. In the design the same checks run **on a schedule**:

- each adapter at its own cadence;
- all of them before the day's first scan, at 03:45 New York.

They feed the health page (ALERTS.md).

Check	Down	Degraded
TradingView scanner: `close > 0`, 50 rows	zero rows; the first row has no usable price; the fetch failed	—
news: every source, for a liquid ticker	no stories from any	some sources failed, with the key named
Alpaca key: the account	rejected (401/403)	—
bar feeds: 40 daily bars each	the fetch failed; no data; under 15 rows; a close NaN or ≤ 0; every volume zero; the last bar over 5 sessions old; out of order or duplicated; a one-day move over 25 %	over 5 % of sessions without volume
indexes ^GSPC, ^IXIC	as bar feeds, with "M cannot be built"	—
cross-source agreement: the only check that catches a confidently wrong feed	the worst close difference over 0.5 %	under 3 shared sessions; fewer than 2 working feeds ("configure a second feed")
RS universe	no sessions; under 1,000 symbols	under 253 sessions
research facts	a required field missing	older than their clock allows, counted in trading sessions (M and groups: not from the last completed session; SIC: 14 days). The old check used 36 hours by the clock, which read every weekend as stale
EDGAR	the ticker map under 5,000; no quarters for a known filer; 403	—
sector map	—	a card sector with no ETF mapping (MARKET.md)

How checks are reported and run:

- Each failing check carries its **fix**.
- Checks run **inside the programs' own environment**, so they never report a different set of keys from the one in use. The first live data check did.
- A red check sends an alert; amber shows on the page.

3.6 Keys

Key	Used by
Alpaca (data)	`alpaca_data`, `alpaca_assets`
each journal account's Alpaca key	`alpaca_account`, per account
Polygon	`polygon`
Finnhub	`finnhub`
the EDGAR User-Agent	`edgar`, `sec_13f`

Where keys live, and what happens without one:

- All keys are in the secret files (BOX.md §4.3).
- A missing key makes its adapter `unavailable` with the reason "key not set", and the health page names it.
- Settings can set and test keys. The old Settings page had no inputs for them (D-T4).

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4. Contracts

It owns	Where
`source_calls`: requests, calls, failures, last error, per family and day	the shared record
`health_checks`: each check's result over time (to build)	the shared record
the bar cache	qp's caches on disk (FEEDS.md)

What it serves. Facts to the desk, the research worker, qp and the journal, through one interface:

- **the call:** `fetch(family, subjects, as_of?)`;
- **each fact returned:** its status, source and the instant it was learned.

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5. When it fails

Event	What happens
a source fails or times out	its families are `unavailable` with the reason; cards still build; the health check goes red and alerts with the fix
a source answers nonsense	the adapter's sanity checks refuse it; the cross-source check catches a plausible but wrong feed
TradingView changes or drops a column	the alignment check refuses the batch; the scan report says why; health goes red; no card gets shifted values
a free tier's limit is hit	the adapter stops at its limit and carries the rest; the limit is named on the health page
a key is missing or refused	the adapter is `unavailable · key not set` (or refused); nothing crashes

6. Built on

The old desk:

- DATA-PIPELINE: every call the card pipeline makes, with endpoints and fallbacks;
- DATA-FEEDS: qp's bar feeds;
- DATACHECK: judging each source by its content;
- CANSLIM §3.1 and §10.4: the free official sources for O'Neil data;
- ACCOUNTS: the journal's Alpaca calls.

What failed there:

ID	Fault	Prevented by
—	the same ticker's facts fetched once per tool	facts fetched once, shared
D-S1	short interest and borrow wiped on a re-quote	each family writes only itself
D-S2, F-14	a missing value read as a number or as bearish	the honest null
D-S3	a column shift misaligning every field	the alignment check
Q-N6	routes with their own Polygon default	no per-route defaults
—	"alpaca ok" printed daily while the key was refused	health judged by content, on a schedule
—	the data check ran only when pressed, and once from the wrong environment	scheduled, inside the programs' environment
D-T4	the fix lines point to a Settings section with no inputs	keys set and tested in Settings
—	a full disk from an unbounded bar cache	the cache's limits

7. Status

Part	Status
the calls themselves, their fallbacks and the alignment check	**built** (legacy, inside the pipeline's stages)
short interest, borrow and news through the fact store, with call counts	**built, in shadow** (step 5)
one adapter per source for every fetch, with fixtures	**partly**; the rest **to build** as each family moves (step D)
the scheduled health checks and the health page	**to build** (step E)
keys set and tested from Settings	**to build**